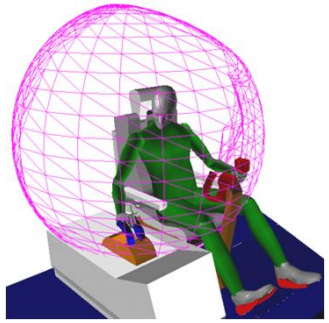
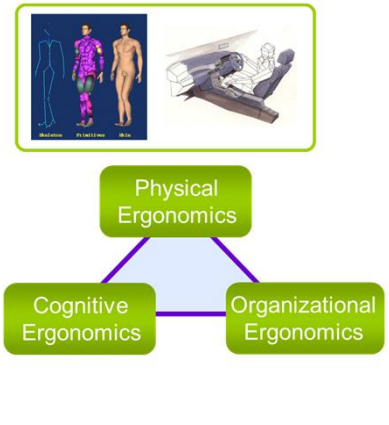
Introduction to
Human Factors & Ergonomics (HF/E)





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Learning Goals

GOALS

- Explain the **three levels of ergonomic design** (corrective, preventative, creative) in relation to HF/E goals.
- Explain **human capabilities and limitations** in sensory, cognitive, and physical domains relevant to design.
- Analyze the **cost-benefit relationship** of ergonomic interventions based on the **Oxenburgh model** and justify investments using ROI concepts.
- Explain the **four basic functions** typically performed by a human or a machine in a **human-machine system**.
- Analyze how **human roles** shift across different **system types (manual, mechanical, automated)** and understand the differences between **open-loop and closed-loop systems**.

- Explain the **advantage of redundancy** to increase the overall reliability of a human-machine system.
- Outline the **generic steps of the systems approach** in HF/E research and apply them to a real-world ergonomic problem under consideration.
- Discuss the **benefits and requirements of a participatory approach** to system design.
- Recognize major **international (ISO) and industry-specific standards and guidelines** essential for HF/E practice.

Three Levels of Ergonomic Design

Evolution from fixing problems to innovating solutions

Corrective (Reactive)

Fixing existing problems or malfunctions in current systems.

When to use: Post-implementation, after accidents/complaints occur.

Preventative (Proactive)

Anticipating potential issues during the design phase to avoid future problems.

When to use: Early design stages, prototype evaluation.

Creative

Innovating new solutions that redefine tasks or systems for optimal performance.

When to use: Conceptual phase, disruptive innovation.

Human Capabilities & Limitations

Why it Matters

The fundamental premise of HF/E is that design must respect human limitations. When system demands exceed human capabilities, performance failures occur.

Sensory Capabilities

Understanding limits of vision, audition, and touch ensures signals are detected. Design must account for thresholds, acuity, and environmental interference.

Cognitive Capabilities

Attention, memory, decision-making, and reaction time have strict limits. Overloading these resources causes errors and mental fatigue.

Physical Capabilities

Anthropometry (body size), biomechanics (strength/ROM), and physiology (endurance) dictate physical design. Mismatches lead to injury and fatigue.

The "Capabilities Stack"

Sensory Capabilities
Input Processing

↓

Cognitive Capabilities
Central Processing

↓

Physical Capabilities
Action Execution

Demand > Capability ⇒ Error

References: Sanders & McCormick (1993); Wickens et al. (2021)

Sensory Capabilities: Vision

Key Visual Factors for Design

Visual Acuity

The ability to resolve fine details. Influenced by distance, illumination, and contrast. Standard acuity is 20/20, meaning resolving 1 min of arc.

Contrast Sensitivity

Ability to distinguish an object from its background. Low contrast reduces legibility and increases error rates, especially in poor lighting.

Color Vision

Approx. 8% of males have color vision deficiency (color blindness). Critical for coding systems (e.g., Red/Green status indicators).

Luminance & Glare

Luminance is the amount of light emitted/reflected. Glare (direct or reflected) causes discomfort and disability, reducing performance.

Design Implications

Text Size (Standard Viewing)	**Min 16-22 min of arc**
Contrast Ratio (ISO)	**Min 3:1 (Text:Background)**
Avoid Blue for Text	**Poor focus (short wavelength)**

Best Practices

- Use redundant coding (Color + Shape/Text)
- Ensure adequate illumination for task type
- Matte finishes to reduce glare
- High contrast (Black on White best)

Visible Spectrum

Human visible range: 380 to 750 nm

750 nm — 555 nm — 380 nm

Photopic vision (daylight) peaks at 555 nm (Green)

References: ISO 9241-303 (2011); Sanders & McCormick (1993)

Sensory Capabilities: Audition & Touch

Auditory System

Hearing Thresholds: Frequency range 20Hz–20kHz; most sensitive at 1–4kHz (speech).

Masking & Alarms: Background noise can mask signals; alarms must be 15dB above ambient noise (ISO 7731).

Tactile & Haptic System

Tactile Sensitivity: Highest in fingertips, lips; critical for texture/shape discrimination.

Haptic Feedback: Vibration/force cues confirm actions (e.g., smartphone buzz, steering wheel vibration).

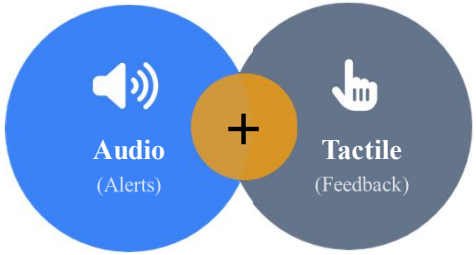
Design Strategy: Multimodal Redundancy

Combine Visual + Auditory + Tactile cues (e.g., a warning that flashes, beeps, and vibrates) to maximize detection probability, especially in high-workload environments.

Multimodal Interaction

Benefit of Redundancy

Reduces reaction time and increases error detection when visual channel is overloaded.



References: ISO 7731 (2003); Wickens et al. (2021)

Cognitive Capabilities: Attention, Memory, Decision

The Cognitive Bottleneck

Human information processing is limited by capacity, speed, and reliability. Design must account for these cognitive constraints to prevent overload and errors.

Attention Types

Selective Attention: Focusing on one source while ignoring others (cocktail party effect).
Divided Attention: Processing multiple tasks simultaneously (limited resource pool).

Working Memory Limits

Limited capacity (Miller's "Magical Number 7 ± 2"). Information decays rapidly (10-20 sec) without rehearsal. Vulnerable to interference.

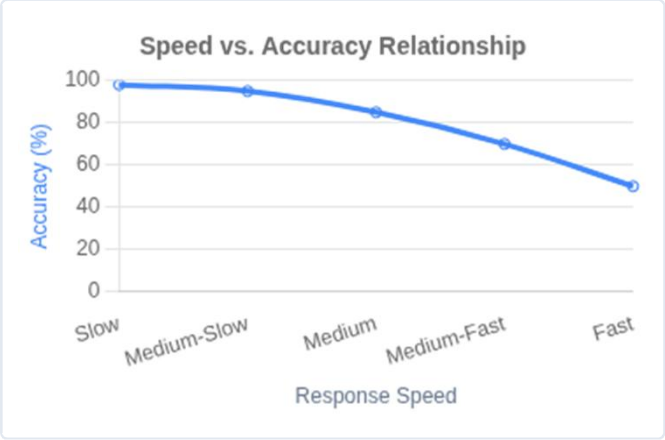
Decision Making & Heuristics

Humans use mental shortcuts (heuristics) under time pressure, leading to biases (e.g., confirmation bias, availability heuristic). System design should support rational choice.

Speed-Accuracy Trade-off (Fitts' Law)

"Faster responses typically lead to higher error rates unless task difficulty is reduced."

Reaction Time **vs.** Error Rate



References: Fitts (1954); Wickens et al. (2021)

Physical Capabilities: Anthropometry & Strength

Key physical data considerations for designing "fit" (P5–P95 variability)

Capability Dimension	Key Concept/Metric	Design Application/ Data Range	Design Strategy
Anthropometry	Body dimensions (static & dynamic) Percentiles (5th, 50th, 95th)	Stature (Height): **P5 ♀ 150cm** to **P95 ♂ 188cm** Eye Height: Critical for sightlines	**Design for Adjustable Range** (e.g., chairs, monitors)
Reach Envelopes	Functional reach area (Primary vs. Secondary) Zone of Convenient Reach	Forward Reach: **50 ~ 60 cm** (comfort zone) Max Reach: **70 ~ 80 cm** (infrequent use)	**Design for the Smallest (P5)** Keep frequently used items in primary zone
Clearances	Space required for body parts access Headroom, Legroom, Maintenance access	Doorway Height: **> 200 cm** Knee Clearance: **60 ~ 70 cm** height	**Design for the Largest (P95/P99)** Ensure accessibility for larger individuals
Strength	Force capability (Isometric, Isotonic) Grip strength, Lift capacity	Max Lift (NIOSH): **23 kg** (optimal conditions) Grip Strength: **♂ 50kg / ♀ 30kg** (varies widely)	**Design for the Weakest** (ensure usability) Minimize force requirements; use mechanical aids
ROM & Endurance	Range of Motion (Joint angles) Fatigue limits over time	Neutral Postures: Keep joints near mid-range Recovery Time: Needed after exertion >15% MVC	**Avoid static loading** (allow postural change) Keep work within comfort angles

Cost-Benefit of Ergonomics: Why Invest?

The Business Case

Ergonomics is not just a health initiative; it is a strategic business investment. Effective HF/E interventions yield substantial returns by reducing losses and enhancing productivity.

Direct Costs of Poor Ergonomics

Visible expenses directly associated with workplace injuries: Medical bills, workers' compensation payments, and legal fees. These are often just the "tip of the iceberg."

Indirect Costs (The Hidden Burden)

Often 4 to 10 times higher than direct costs: Absenteeism, turnover, training replacement workers, accident investigation time, lost productivity, and lower morale.

Strategic Benefits of Investment

Beyond cost avoidance: Improved product quality, higher efficiency/throughput, better employee engagement, and enhanced corporate reputation.

The "Iceberg Effect" of Costs

Total Injury Cost

= Direct Costs + Indirect Costs

Direct Costs $1

Medical Expenses / Worker's Compensation

Indirect Costs $4 ~ $10

Hidden Costs (4~10x Direct Costs)

Typical ROI

1 : 3+

For every $1 invested, $3+ returned

References: OSHA (2019); Hendrick (1996)

Cost-Benefit of Ergonomics: Why Invest?

The economic impact of neglecting human factors in system design manifests in various direct and indirect costs.

Cost Category	Description & Impact	Industry Example
Injury & Illness	Workers' compensation claims, medical expenses for MSDs (Musculoskeletal Disorders)	**Manufacturing:** Back injuries from heavy lifting without assist devices ($40k-$80k per claim)
Absenteeism	Lost workdays due to pain/fatigue, need for temporary replacement workers	**Logistics:** High sick leave rates in warehouse staff due to repetitive bending/reaching
High Turnover	Costs of recruiting and training new staff; loss of experienced workers	**Service/Call Centers:** Burnout from poor workstation setup leading to 30%+ annual turnover
Quality Defects	Errors caused by fatigue, poor lighting, or difficult assembly processes (scrap/rework)	**Electronics:** Assembly errors increase by 20% in last 2 hours of shift due to visual fatigue
Operational Downtime	Line stoppages due to accidents or slower work pace from awkward postures	**Automotive:** Assembly line halts when workers struggle with poor tool accessibility
Liability & Compliance	OSHA fines, legal fees, regulatory penalties for unsafe conditions	**Construction:** Citations for failing to provide ergonomic tools for vibration hazards

Reference: OSHA (2019)

Benefits and ROI: Case Evidence

Real-world examples demonstrating the tangible value of ergonomic interventions

Manufacturing Assembly Line

Redesigned workstations to minimize reaching distances and introduced lift-assist devices for components over 15kg.

50%
Reduction in Back Injuries

15%
Productivity Increase

Office Ergonomics Program

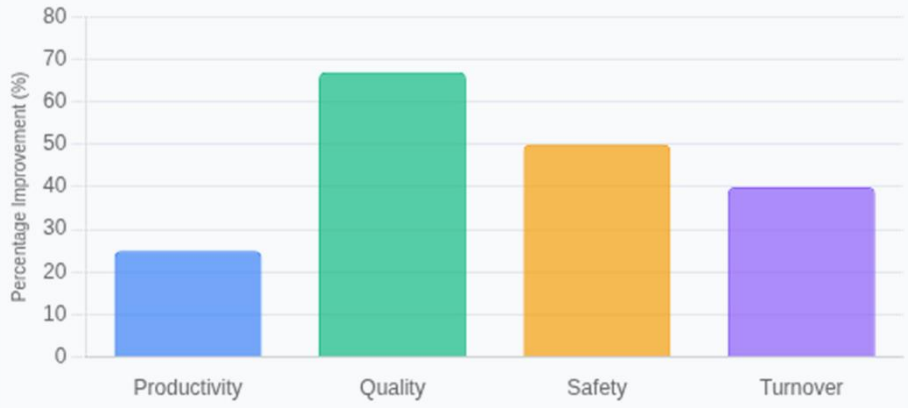
Implemented adjustable furniture, monitor arms, and comprehensive employee training on posture and breaks.

$60,000
Workers' Comp Savings

25%
Absenteeism Drop

Impact Distribution

Average improvements across industries



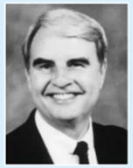
*"**Good Ergonomics** is **Good Economics**. The cost of prevention is significantly lower than the cost of injury and lost productivity."*

- Hendrick (1996)

Good Ergonomics = Good Economics

Good Ergonomics Is Good Economics

By
Hal W. Hendrick
1996 HFES Presidential Address



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INTRODUCTION

One of the clearest ways to delineate a discipline is by its unique technology. At its recent workshop, the HFES Strategic Planning Task Force noted, as have others internationally, that the technology of human factors/ergonomics is human-system interface technology. Thus, the discipline of human factors can be defined as the development and application of human-system interface technology.

Human-system interface technology deals with the interfaces between humans and the other system components, including hardware, software, environments, jobs, and organizational structures and processes. Like the technology of other design-related disciplines, it includes specifications, guidelines, methods, and tools. As noted by the Strategic Planning Task Force, we use our discipline's technology for improving the quality of life, including health, safety, comfort, usability, and productivity. As a science we study human capabilities, limitations, and other characteristics for the purpose of developing human-system interface technology. As a practice, we apply human-system interface technology to the analysis, design, evaluation, standardization, and control of systems. It is this technology that clearly defines us as a unique, stand-alone discipline, that identifies who we are, what we do, and what we offer for the betterment of society.

Although they may come from a variety of professional backgrounds, such as psychology, engineering, safety, the rehabilitation professions, or medicine, it is their professional education and training in human-system interface technology that qualifies persons as human factors/ergonomics professionals. Indeed, the discipline needs both the breadth and richness of these professional backgrounds as well as the education and training in the unique technology of human factors/ergonomics.

Human factors/ergonomics professionals have long recognized the tremendous potential of our discipline for improving people's health, safety, and comfort and both human and system productivity. Indeed, through the application of our unique human-system interface technology, we have the potential to truly make a differ-

Tractor-trailer, before.

Tractor-trailer, after.

C-141 aircraft.

total investment of $6,900, a hard cost savings of $65,000 per year was achieved – a 1 to 9.4 cost-benefit ratio (Warkotsch, 1994).

Other innovations. Other innovations by this same collaborative effort between Stellenbosch University, Ergotech, and various forestry companies have included (a) the development of a unique, lightweight, environmentally friendly pipe type of timber chute for more efficient and safe transporting of logs down slopes; (b) redesign of three-wheeled hydrostatic loaders to reduce both excessive whole-body vibration and noise; (c) classifying different terrain conditions – including ground slope, roughness, and other conditions – and determining the most effective tree harvesting system (method and equipment) for each; and (d) developing ergonomic checklists and work environment surveys tailored to the forest industry. All are expected to result in significant cost savings, as well as greater employee satisfaction and improved quality of work life (Warkotsch, 1994).

I believe this is a good example of what ergonomics potentially can contribute to any given industry when there is a true collaborative effort and commitment.

C-141 Transport Aircraft

Some 35 years ago, I joined the U.S. Air Force's C-141 aircraft development system program office as the project engineer for both human factors and the alternate mission provisions. The C-141 was to be designed so that its cargo compartment, through the installation of alternate mission kits, could be reconfigured for cargo aerial delivery, carrying paratroopers and paratroop jumping, carrying passengers, or medical evacuation. As initially configured, anything that did not absolutely have to be included in the aircraft for straight cargo carrying was placed in one of the alternate mission kits, making them heavy and complex and requiring considerable time and effort to install.

By meeting with the intended user organization, the Air Force Material Air Transport Command, and discussing their organizational design and management plan for actual utilization of the aircraft, I was able to identify numerous kit components that rarely ever would be removed from the airplane. Using these data, I worked with the Lockheed design engineers to reconfigure the kits to remove these components and, instead, install them permanently in the aircraft. As documented by the engineering change proposals, this effort greatly simplified the system and reduced actual operational aircraft weight and, thus, related operating and maintenance costs for more than 200 aircraft over the past 35 years. The changes also reduced installation time and labor and storage requirements for the kits. In addition, it saved over $2 million in the initial cost of the aircraft fleet. I believe this is a good illustration of how macroergonomic considerations can result in highly cost-effective microergonomic design improvements to systems.

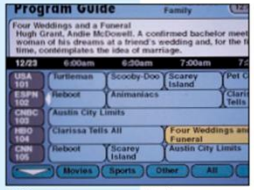
These and numerous other cost-benefit human factors evaluations and improvements to the C-141's design came at a total cost of less than $500,000 of professional human factors effort and resulted

Replacement for forklift truck lines.

Forklift truck, new ergonomic design.

TV and VCR remote controls, before.

Left: Redesigned System Link remote control; right: DSS system remote control.



DSS system, on-screen display.

Product Design or Redesign

The economic benefit of ergonomic design or redesign of a product can be assessed in several ways – for example, by its impact on (a) the value of the company's stock, (b) sales, (c) productivity, or (d) reductions in accidents. Four very different kinds of products are provided here as illustrations of each of these beneficial economic impacts.

Replacement for forklift truck lines. Alan Hedge and his colleagues at the Human Factors Laboratory at Cornell University participated with Pelican Design, a New York industrial design company, and the Raymond Corporation in the design and development of a new generation of forklift trucks to replace Raymond's two existing product lines. Human factors design principles were given prime consideration, and an "inside-out" human-centered approach was taken, with the form of the truck being built around the operator's needs. The goal was to maximize operator comfort, minimize accident risks, and maximize productivity by optimizing task cycle times. At the time the development project was begun, Raymond's market share had eroded from its former position of dominance in the market of over 70% of sales to about 30%, and it was shrinking.

The new narrow isle and swing-reach truck lines were introduced in the United States in 1992, and the swing-reach line was introduced in Europe in 1993. Order books at Raymond are full, and the company is once again enjoying success. Raymond stock has risen from around $6 per share at the start of the project to around $21 today (Alan Hedge, personal communication).

TV and VCR remote controls. Thomson Consumer Electronics first developed its highly successful approach to user-centered design when it developed System Link, an ergonomically oriented remote control that can operate various types of products made by different manufacturers. The original Thomson remote control design differed little from the competition's: a rectangular box with rows of small, identical buttons. It is the one in the middle of the picture at left ("before").

Using the company's user-centered design approach, the initial design was replaced with the new ergonomic one, shown on the left in the "after" picture, which, among other things, was easier to grasp, used color-coded, soft-touch rubber buttons in distinctive sizes and shapes, and separated the VCR and TV buttons above and below the keypad. When introduced in 1988, this new, ergonomically designed System Link remote control gained the jump on the competition, and Thomson has since sold millions of them. As a result of this success, user-centered ergonomic design has become a key aspect of all new Thomson development projects (March, 1994).

DSS system. A more recent highly successful example is Thomson's RCA DSS satellite digital television system. All aspects, including the on-screen display and remote control, utilized user-centered design and received extensive ergonomic attention (March, 1994). These units now are selling like hotcakes.

Example: ROI Calculation

Return On Investment (ROI)

ROI = (Benefits - Costs) / Costs × 100%

1 Total Intervention Costs

Equipment (Workstations & Lift Assists):	**$15,000**
Training ($3k) + Install Downtime ($2k):	**$5,000**
Total Costs (C):	**$20,000**

2 Annual Benefits (Savings)

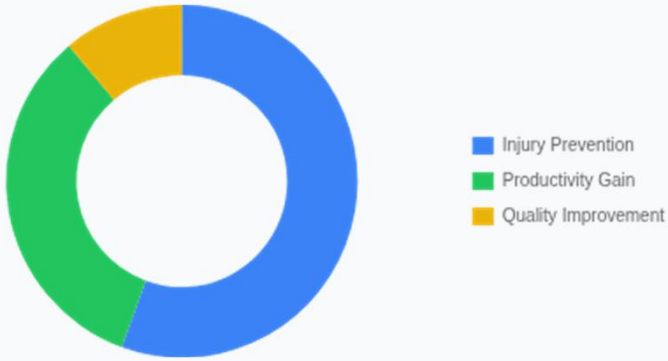
Injury Prevention (Workers' Compensation):	**$25,000**
Productivity (5% faster) + Quality:	**$20,000**
Total Benefits (B):	**$45,000**

First Year ROI

125%

Payback Period (= Costs/Benefits): 5.3 months

Benefit Breakdown



Sensitivity Analysis (Scenarios)

↓ Worst Case (Benefits -20%)	**ROI = 80%**
↑ Cost Overrun (Costs +20%)	**ROI = 87.5%**
Safety Only (No Productivity Gain)	**ROI = 25%**

Project viable even without productivity gains.

Oxenburgh Model Application

Automotive parts assembly line case study demonstrating comprehensive cost-benefit analysis.

Problem Context

High musculoskeletal disorder incidence rate (12 cases/year) due to poor workstation height, repetitive reaching, and heavy lifting without aids.

Pre-Intervention Costs

$180,000 Annual Injury Costs including workers' compensation, medical expenses, lost productivity, and replacement training.

Oxenburgh Model Calculation

Direct Costs (worker's compensation/medical expenses)	$36,000
Indirect Costs (4:1 Ratio)	$144,000
(Productivity loss, training, defects)	
Total Annual Cost Impact	**$180,000**

Ergonomic Intervention

Height-adjustable workstations, mechanical lift assists, improved lighting, and job rotation implementation.

Initial Investment: $85,000

-75% Injury Reduction (3 cases)	**+8%** Productivity Increase
-40% Quality Defects	**$215k** Total Annual Savings

Financial Impact Analysis

System: Definition & Characteristics

System Definition

An entity that exists to carry out a specific **purpose**. It is composed of humans, machines, and other **elements** that **interact** to **accomplish a goal that these components could Not achieve independently**.

References: Bailey (1982); Sanders & McCormick (1993)

Goals

Systems have defined goals. Improvements must be evaluated at the total system level.

Hierarchical

Systems consist of subsystems and components. Analysis requires setting a limit of resolution.

Operating in Environment

From immediate workspace to general environment. Constraints shape design and use.

Components Serving Functions

Sensing, Information Storage, Processing & Decision, and Action.

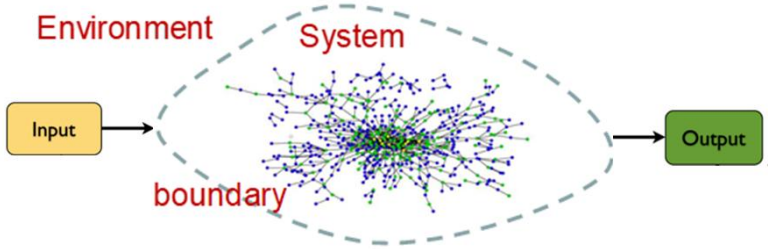
Interaction between Components

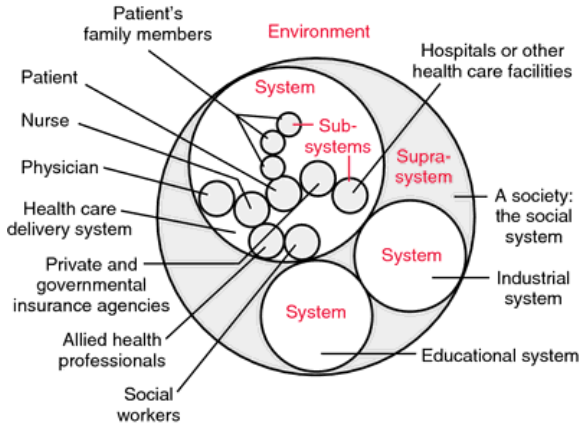
Inputs/outputs link elements. Structural and functional relationships between components determine performance & reliability.

System Boundary

Explicitly defines what is inside vs. outside the system and corresponding inputs and outputs for analysis scope.

System Structure & Interaction





System Types & HF/E Implications

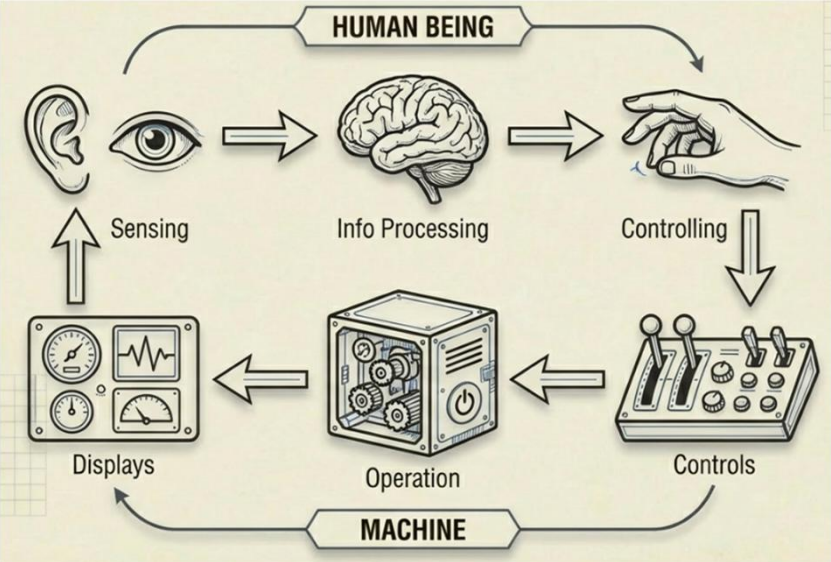
Different levels of automation require distinct HF/E considerations, but human involvement remains critical across all system types.

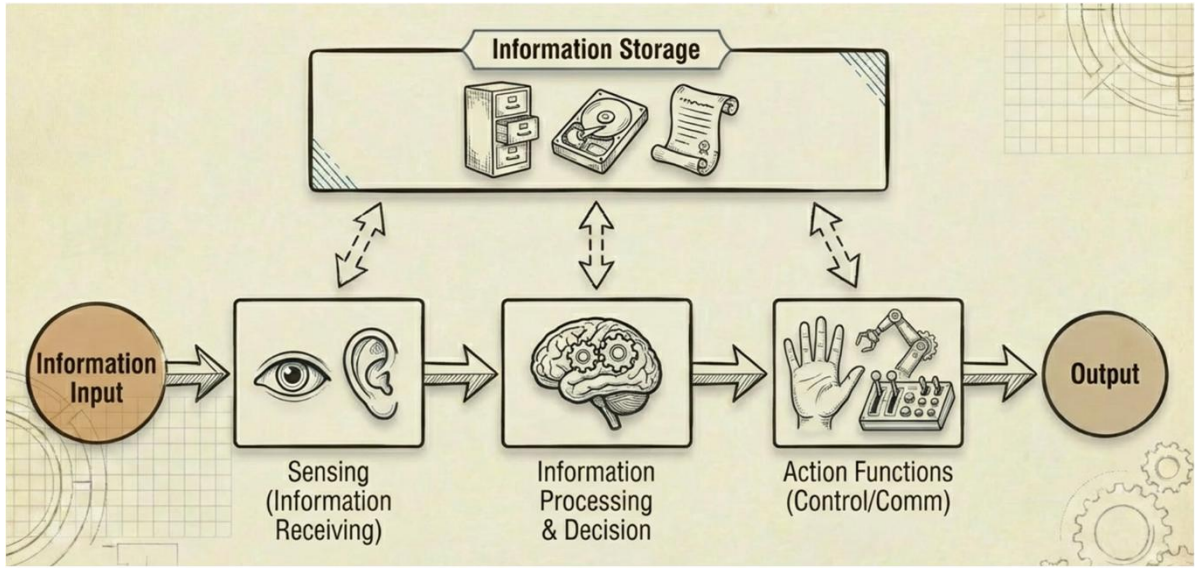
System Type	Examples	Human's Role	System's Role	Primary HF/E Focus
Manual System	hammering, shoveling	Provides power Executes control Makes all decisions	Passive tool Extends capability No autonomy	Physical workload, tool design, posture, biomechanics, force requirements
Mechanical System (Semi-Automatic)	driving a car, operating a lathe	Initiates/supervises Adjusts controls Monitors process	Provides power Programmed functions Feedback display	Control/display design, feedback adequacy, guarding, mode confusion.
Automated System	autonomous vehicle, CNC	Programs/Configures Monitors status Handles exceptions	Autonomous operation Self-regulation Complex sequences	Supervisory control, mode awareness, trust, failure recovery, control transfer

Key Insight: Automation does **NOT eliminate human factors needs**; it **shifts** the human role **from active control to monitoring & exception handling**, creating new cognitive challenges.

Four Basic Functions in Human-Machine Systems

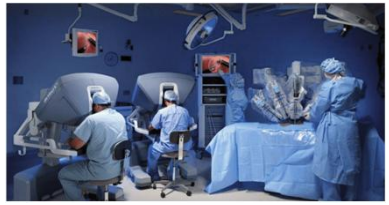
Every component of human or machine performs one or more of four basic functions (**sensing, information processing & decision, information storge, and action**) to achieve system goals.





- ❑ Human-machine system
- ❑ Human-product system
- ❑ Human-computer system

- ❑ Human-vehicle system
- ❑ Human-robot system
- ❑ Human-automation system



Open-Loop vs. Closed-Loop Systems

Open-Loop System

Characteristics: Activated once, no further control possible during operation.

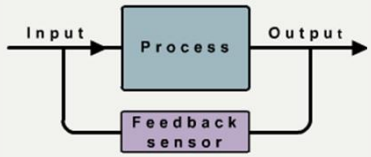
Feedback: No operative feedback during execution. Feedback only informs subsequent attempts.

Examples: Unguided rocket, fixed-time washing machine, single-shot stamping press.

Human Role: Initiate-and-leave. Setting initial parameters correctly is critical.

Design Implication: Requires robust initial settings, high tolerances, and fail-safe mechanisms.

Closed-Loop System



Characteristics: Continuous control; system state is continuously measured and adjusted.

Feedback: Uses feedback to detect error (desired - actual) and correct to maintain state.

Examples: Automobile driving, thermostat, chemical process control, homeostasis.

Human Role: Continuous monitoring or supervisory control with interventions.

Design Implication: Needs timely, reliable feedback channels and stable control dynamics.

System Reliability: Series vs Parallel

Parallel System Formula (Redundancy)

$$R_{system} = 1 - F_1 \times F_2 \times \ldots \times F_n = 1 - (1 - R_1) \times (1 - R_2) \ldots \times (1 - R_n)$$

1 Component Reliability (R_i)

Component A (Primary):	$R_A = 0.70$
Component B (Backup/Redundant):	$R_B = 0.70$

Assumption: Failures are independent events

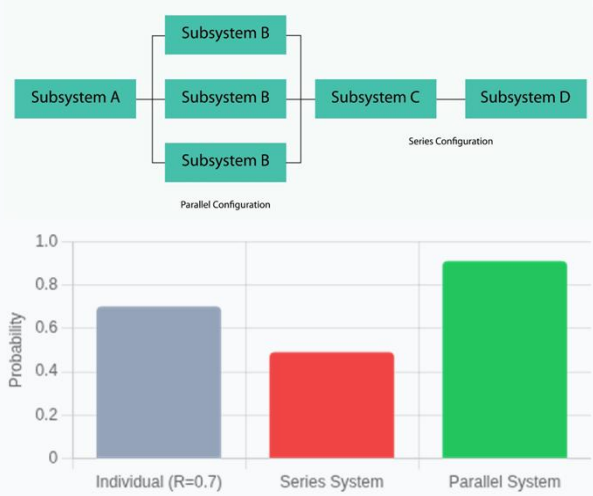
2 Series System (Weakest Link)

Formula:	$R_{series} = R_A \times R_B$
Calculation:	$R_{series} = 0.70 \times 0.70$
Result:	$R_{series} = 0.49$

3 Parallel System (Redundancy) Target

Probability of Failure ($F_i = 1 - R_i$):	$F_i = 0.30$
System Failure ($F_{system} = F_A \times F_B$):	$F_{parallel} = 0.30 \times 0.30 = 0.09$
System Reliability ($1 - F_{system}$):	$R_{parallel} = 1 = F_{parallel} = 1 - 0.09 = 0.91$

Impact of Redundancy



Design Implications

Critical Functions	Use **parallel redundancy** for **life-critical or high-cost failure modes**.
Weakest Link	In series systems, overall reliability is always less than the worst component.
Human Role	Humans often serve as **redundant monitors to automated systems**.

Generic Steps in HF/E Projects

Standard human-centered design process aligned with ISO 9241-210

1. **Understand & Specify Context of Use**
   Identify users, tasks, environment, and stakeholder needs.

2. **Specify User Requirements**
   Establish ergonomic criteria and performance goals based on user needs.

3. **Produce Design Solutions**
   Create concepts, prototypes, and specifications (Iterative loop starts).

4. **Evaluate Designs Against Requirements**
   User testing, expert reviews, and validation against criteria.

5. **Iterate or Proceed to Implementation**
   Refine design if requirements unmet; otherwise, finalize system.

6. **Deploy & Monitor**
   Implementation in real world with ongoing feedback and maintenance.

Reference: ISO 9241-210 (2019). Ergonomics of Human-System Interaction — Part 210: Human-Centred Design for Interactive Systems

Participatory Approach: Benefits and Requirements

Key Benefits

- **Higher Acceptance:** Employees are more likely to embrace changes they helped design.
- **Sustained Gains:** Performance improvements persist longer compared to top-down mandates.
- **Smoother Change Management:** Reduced resistance and frustration during implementation.
- **Better Solutions:** Leveraging frontline expertise leads to more practical and effective designs.

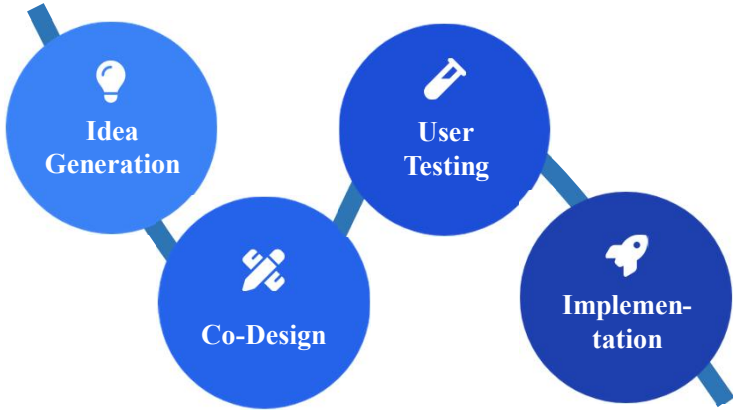
Essential Requirements

- **Early Involvement:** Engage users from the initial concept phase, not just at the end.
- **Representative Participation:** Include diverse voices from all affected user groups.
- **Transparent Communication:** Clear goals, constraints, and feedback loops.

Why Trust Matters

Trust and involvement are critical for aligning with sociotechnical systems theory. Joint optimization of social and technical systems requires mutual respect and shared ownership of the design process.

Iterative Participatory Cycle



Historical Evidence

"Coch & French (1948) demonstrated that production fell in control groups after change, but rose immediately and was maintained in groups with total participation, proving the value of employee involvement."

Why Standards & Guidelines Matter

The Role of Standards

Standards provide established criteria, methods, and practices to ensure systems are safe, reliable, and usable across different contexts and users.

Ensure Safety & Minimize Risk

Prevent accidents by adhering to proven safety limits (e.g., lifting limits, exposure thresholds).

Guarantee Usability & Performance

Ensure systems are effective, efficient, and satisfying for specified users to achieve specified goals.

Facilitate Interoperability

Enable different systems and components to work together seamlessly (e.g., consistent control layouts).

Regulatory Compliance

Meet legal requirements (OSHA, EU Directives) to avoid liability and market access barriers.

Strategic Impact

Consistent Design
Reduces design guesswork and variability across products.

Global Market Access
Meets international requirements for export (ISO/IEC).

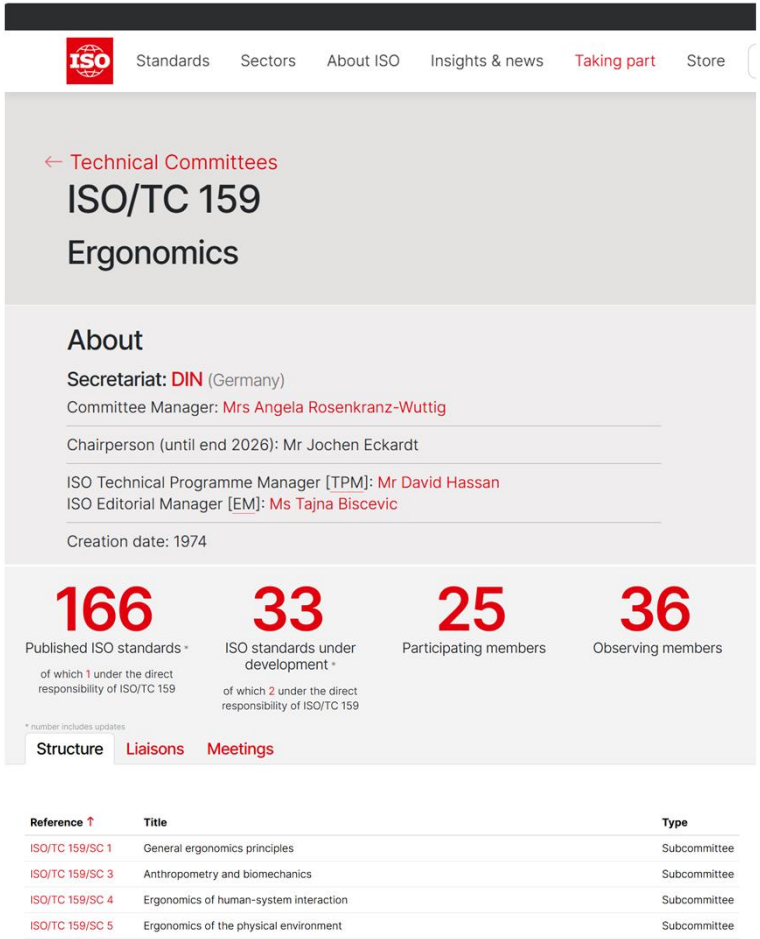
Cost Reduction
Avoids costly redesigns and post-market fixes.

User Trust
Builds confidence through predictable interactions.

Key ISO Standards in HF/E

Fundamental International Organization for Standardization (ISO) documents guiding ergonomic practice

Standard	*Year published*	*Title*
General		
ISO 7317	1987	Ergonomics: Standards Guidelines for Designers
ISO 6385	1981	Ergonomic Principles in the Design of Worksystems
ISO 18529	2000	Ergonomics – Ergonomics of Human–System Interaction – Human-Centred Lifecycle Process Descriptions
ISO 13407	1999	Human-Centred Design Processes for Interactive Systems
Physical workload		
ISO 1128-1-3	2000	Ergonomics – Manual Handling Parts 1 to 3
ISO 53120	1996	Equipment for Manual Handling
ISO 11226	2000	Ergonomics – Evaluation of Static Work Postures
Physical environment		
ISO 13340-1	1995	Protective Equipment in General
ISO 13340	1996	Protective Gloves
Workspace design		
ISO 9241	1990s	Ergonomic Requirements for Office Work with Visual Display Terminals – Parts 1–9
ISO 11064-1		Ergonomic Design of Control Centres. Control Room Layout
Physical work/stress		
ISO 8996	1990	Ergonomics – Determination of Metabolic Heat Production
Mental work/HCI		
ISO 9241	1990s	Ergonomic Requirements for Office Work with Visual Display Terminals – Parts 10–17
Displays and controls		
ISO 9355	1999	Ergonomic Principles for the Design of Signals
ISO 11429	1996	Ergonomics – System of Auditory and Visual Danger and Information Signals
ISO 7731	1986	Ergonomics. Danger Signals for Public and Work Areas Auditory Danger Signals
Work environment		
ISO 13731	1995	Ergonomics of the Thermal Environment. Vocabulary and Symbols
ISO 11399	2001	Ergonomics of the Thermal Environment – Principles and Application of Relevant International Standards
ISO 9116-1	1989	Lighting in General
ISO 91160-10	1989	Interior Lighting
ISO 8995	1989	Principles of Visual Ergonomics – The Lighting of Indoor Work Systems
ISO 1996-1	1982	Acoustics – Description and Measurement of Environmental Noise
ISO 9921-1	1996	Ergonomic Assessment of Speech Communication
Safety		
ISO 15534-1-3	2000	Ergonomic Design for the Safety of Machinery, Parts 1–3
ISO 14121	1999	Safety of Machinery – Principles of Risk Assessment
ISO 13854	1996	Safety of Machinery – Minimum Gaps to Avoid Crushing Parts of the Human Body



Example: ISO 7250-1

INTERNATIONAL STANDARD **ISO 7250-1**

Second edition
2017-08

Basic human body measurements for technological design —

Part 1:
Body measurement definitions and landmarks

Définitions des mesures de base du corps humain pour la conception technologique —

Partie 1: Définitions des mesures du corps et repères

Reference number
ISO 7250-1:2017(E)

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6.2.6 Shoulder-elbow length

Description: Vertical distance from acromion to the bottom of the elbow bent at a right angle with the forearm horizontal. See Figure 38.

Method: Subject stands, or sits erect with the feet supported so that the femora are horizontal and parallel to each other. Upper arms hang freely downwards and forearms are horizontal.

Instrument: Large sliding caliper.

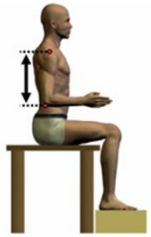

Figure 38 — Shoulder-elbow length

6.2.7 Shoulder (biacromial) breadth

Description: Distance along a straight line from acromion to acromion. See Figure 39.

Method: Subject sits or stands fully erect with shoulders relaxed.

Instrument: Large sliding caliper or large spreading caliper.

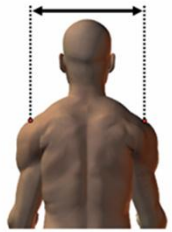

Figure 39 — Shoulder (biacromial) breadth

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Regional & Industry Standards/Guidelines

Key standards governing specific domains and regional applications in HF/E.

Standard ID	Scope & Description
ANSI/HFES 100 USA / Office	**Human Factors Engineering of Computer Workstations** Specifies requirements for the design of hardware (input devices, displays) and furniture (chairs, desks) for computer workstations to maximize comfort and performance.
MIL-STD-1472H Defense / DoD	**Department of Defense Design Criteria Standard: Human Engineering** Comprehensive criteria for military systems, equipment, and facilities. Covers everything from control layout and environmental conditions to maintainability and anthropometry.
IEC 62366 Medical Devices	**Application of Usability Engineering to Medical Devices** Specifies a process for a manufacturer to analyze, specify, develop and evaluate the usability of a medical device as it relates to safety. Critical for FDA/CE approval.
DOT/FAA/AM-24/23 Aerospace Industry	**FAA Guides for Human Factors Considerations in the Design and Evaluation of Flight Deck Displays and Controls** Provides regulatory requirements and guidance for the design and evaluation of flight deck displays and controls.

Example: MIL-STD-1472H

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METRIC
MIL-STD-1472H
15 September 2020
SUPERSEDING
MIL-STD-1472G
w/CHANGE 1
17 January 2019

DEPARTMENT OF DEFENSE
DESIGN CRITERIA STANDARD
HUMAN ENGINEERING

AMSC N/A AREA HFAC

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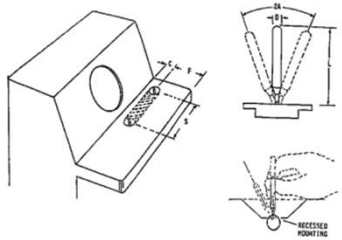
5.1.3.5.2.5 Dimensions and clearance. Joystick dimensions and clearance shall be in accordance with the criteria on figure 5.



	Dimensions		Resistance	A, Displacement	Clearance		
	D, Diameter	L, Length			S, Display CL to Stick CL	C, Around Stick	F, Stick CL to Shelf Front
Minimum	6.5 mm (0.25 in)	75 mm (3 in)	3.3 N (12 oz)		0	1/	120 mm (4.75 in)
Maximum	16 mm (0.625 in)	150 mm (6 in)	8.9 N (32 oz)	45° (785 mrad)	400 mm (15.75 in)		250 mm (10 in)

NOTE:
1/ Maximum stick excursion plus 100 mm (4 in).

FIGURE 5. Displacement joysticks.

5.1.3.5.2.5.1 Hand grip length. The hand grip length shall be 110 to 180 mm (4.3 to 7.1 in).

5.1.3.5.2.5.2 Hand grip diameter. The hand grip diameter shall be not more than 50 mm (2 in).

5.1.3.5.2.5.3 Clearance. Clearances of 100 mm (4 in) to the side and 50 mm (2 in) to the rear shall be provided to allow for hand movement.

5.1.3.5.2.5.4 Full range of motion. Clearance shall be provided to allow for the full range of joystick motion.

5.1.3.5.2.5.5 Actuation of buttons. Clearance shall be provided to enable actuation of joystick buttons while in any position within the joystick range of motion.

5.1.3.5.2.5.6 Mounting. Joysticks shall be mounted to provide forearm or wrist support.

5.1.3.5.2.5.7 Modular devices. Modular devices shall be mounted to allow actuation of the joystick without slippage, movement, or tilting of the mounting base.

5.1.3.5.3 Finger-operated displacement joysticks.

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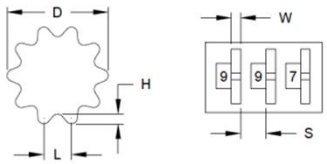

	D, Diameter	L, Trough Distance	W, Width	H, Depth	S, Separation	Resistance
Minimum	29 mm (1.125 in)	11 mm (0.43 in)	3 mm (0.125 in)	3 mm (0.125 in)	10 mm (0.4 in)	1.7 N (6 oz)
Maximum	75 mm (3 in)	19 mm (0.75 in)	--	6 mm (0.25 in)	--	5.6 N (20 oz)

FIGURE 8. Discrete thumbwheel control.

5.1.4.1.1.3.11 Resistance. Detents shall be provided for discrete position thumbwheels.

5.1.4.1.1.3.11.1 Elastic. Resistance shall be elastic, build up, and then decrease as each detent is approached so that the control snaps into position without stopping between adjacent detents.

5.1.4.1.1.3.11.2 Separation. The separation between adjacent edges of thumbwheel controls shall conform to the criteria on figure 8. Separation shall be sufficient to preclude accidental actuation of adjacent controls during normal control operation.

5.1.4.1.2 Continuous adjustment rotary controls.

5.1.4.1.2.1 Knobs. Knob design shall meet the criteria in 5.1.4.1.2.1.1 through 5.1.4.1.2.1.7.3.

5.1.4.1.2.1.1 Use. Knobs should be used when low forces or precise adjustments of a continuous variable are required. For most tasks, a moving knob with fixed scale should be used instead of a moving scale with fixed index. Special knob designs consisting of a bar-type knob and open-window skirt may be used for applications in which visual confusion may be reduced by exposing only one number of a scale at a time. Handle-like rotary controls may be used for special applications (i.e., a motorcycle accelerator control).

5.1.4.1.2.1.2 Single revolution controls. If positions of single revolution controls must be distinguished, a pointer or marker shall be available on the knob.

5.1.4.1.2.1.3 Shapes. Knobs that perform the same function shall have the same shape.

5.1.4.1.2.1.3.1 Knob function. Knob shape shall be determined by the knob's function and use.

5.1.4.1.2.1.3.2 Representative knob shapes. Representative knob shapes are shown on figure 9.

Example: DOT/FAA/AM-24/23

Human Factors Considerations in the Design and Evaluation of Flight Deck Displays and Controls

Version 3.0

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Final Report – September 2024

DOT/FAA/AM-24/23
DOT-VNTSC-FAA-24-10

Prepared for:
Federal Aviation Administration
Human Factors Division
800 Independence Avenue, SW
Washington, DC 20591

Character Size

- The minimum visual angle for characters should be 16 minutes of arc. However, a minimum visual angle of 24 minutes of arc is preferred for characters that must be read under aircraft environmental conditions. (Ahlstrom and Longo, 2003; McAnulty, 1995)
  Note: 1 degree = 60 minutes of arc (Ahlstrom and Longo, 2003; McAnulty, 1995)
- When measured from the greatest anticipated viewing distance, the height of alphanumeric characters and pictorial symbols shall subtend not less than 2.9 milliradians (10 minutes) of visual angle and should subtend not less than 4.5 milliradians (15 minutes) of visual angle. [MIL-STD-1472H, 5.17.18.2]
- Aircraft display characters and symbols that must be read in flight shall subtend not less than 7.0 milliradians (24 minutes) of visual angle. [MIL-STD-1472H, 5.17.18.3]
- Character width shall be approximately 0.9 of the height. [MIL-STD-1472H, 5.17.18.5]
- The width of letters should be ⅗ to ⅘ of the height, except for "I", which should be 1/10 to ⅗ of the height. [MIL-STD-1472H, 5.4.7.6.1]
- The width of numerals should be ⅗ of the height, except for "4", which should be ⅘ of the height, and "1", which should be ⅕ of the height. [MIL-STD-1472H, 5.4.7.6.2]
- The ratio of character height to width should be: (McAnulty, 1995)
  - At least 1:0.7 to 1:0.9 for equally spaced characters and lines of 80 or fewer characters
  - At least 1:0.5 if it is necessary to have more than 80 characters per line, or
  - As much as 1:1 for characters such as "M" and "W" for proportionally spaced characters.
- Stroke width in normal illumination. If labels are expected to be read under normal illumination and black characters are specified on a white or light background, the stroke width should be ⅙ to 1/7 of the height. [MIL-STD-1472H, 5.4.7.4.1]
- Stroke width in dim illumination. Where dark adaptation is required or legibility at night is a critical factor and white characters are specified on a dark background, the stroke width of the characters should be 1/7 to ⅛ of the height. [MIL-STD-1472H, 5.4.7.4.2]
- Stroke width for transilluminated characters. For transilluminated characters, the stroke width should be 1/10 of the height. [MIL-STD-1472H, 5.4.7.4.3]
- Stroke width shall be 0.1667 to 0.125 of numeral or letter height but may be narrower for light markings on a dark background. Stroke width shall be the same for all letters and numerals of equal height. [MIL-STD-1472H, 5.2.2.13.9.2]
- Text that is in color should be presented at a larger size than black text (on a white background) to provide similar legibility. (Cardosi and Murphy, 1995)

Character Spacing

- The minimum spacing between characters or scale graduation marks should be at least one stroke width or 10% of the character height, whichever is greater. (McAnulty, 1995; MIL-STD-1472H, 5.4.7.8; MIL-HDBK-759C, 5.5.5.4)
- The minimum spacing between words should be at least one character width for equally spaced characters (with the exception of "I" or "1") or the width of the capital letter "N" for proportionally spaced characters. (Ahlstrom and Longo, 2003; MIL-HDBK-759C, 5.5.5.5)
- Word spacing. The minimum space between words shall not be less than ⅗ of the character height (MIL-HDBK-759C, 5.4.7.9)
- The minimum spacing between lines in a display should be two stroke widths or 15% of character height, whichever is greater. (McAnulty, 1995)

C.1 Evaluation Procedures for AC 25-11: Flight Deck Displays

This following checklist provides a list of human factors topics to consider during a display system evaluation. The topics are applicable to all display systems and controls. More information about each topic can be found in the body of this document. The design of the checklist and the wording of the items originated from research conducted to develop the Electronic Flight Bag (EFB) User-Interface Assessment Tool (see Chandra and Yeh (2006) for more information). That tool was then adapted to serve as a companion to AC 25-11A.

The display system and/or control should be evaluated with respect to each topic in this checklist. Comments about each item should be provided with supporting examples. It is important to note that comments can be positive, highlighting the good aspects of a display system's design.

DISPLAY CHARACTERISTICS

- Size and resolution
- Luminance, contrast ratio
- Gray scale, chromaticity
- Glare, reflections
- Defects
- Dynamics (refresh rate, update rate and response time)
- Lighting Issues (day vs. night use)
  - Brightness adjustment (automatic or manual)
- Mechanical integrity, durability
- Installation
  - Visual obstructions, cross-flight-deck viewing
  - Viewability and viewing envelope
  - Interaction/interference with other flight deck systems

DISPLAY INFORMATION ELEMENTS

Text

- Size, font style, case, spacing, color, contrast, illumination
- Clarity and consistency of terms and abbreviations

Symbols, Graphical Depictions and Images

- Clarity of intended meaning, confusability, consistency with other flight deck displays
- Position accuracy
- Number, legibility and distinctiveness
- Image size, resolution and line width
- Color and luminance
- Orientation, ownship directionality
- Zooming and panning

DISPLAY INFORMATION ELEMENTS (cont.)

Display Indications, Annunciations, Messages and Alerts

- Ease and accuracy of interpretation
  - Data accuracy, trend depiction
  - Filtering characteristics, depiction of operating range, delimiters
  - Visual clutter, arrangement of information, frame of reference
  - Use of color (especially red and amber/yellow) and color-coding
  - Intensity (aural or visual)

Sharing Information on the Display (Multiple Applications)

- Managing shared displays (overlays, time-sharing, concurrent displays in separate windows)
  - Smooth transitions between displays
  - Accuracy of motion (e.g., position, time)
  - Readability
  - Information prioritization, blending
  - De-cluttering
- Consistency across shared information displays (e.g., symbols/colors, orientation, data range)
- Ease of movement between and within pages
  - Accessing functions, options, and specific information/pages
  - Identifying the current page

Example: MIL-STD-1472H

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METRIC
MIL-STD-1472H
15 September 2020
SUPERSEDING
MIL-STD-1472G
w/CHANGE 1
17 January 2019

DEPARTMENT OF DEFENSE
DESIGN CRITERIA STANDARD
HUMAN ENGINEERING

AMSC N/A AREA HFAC

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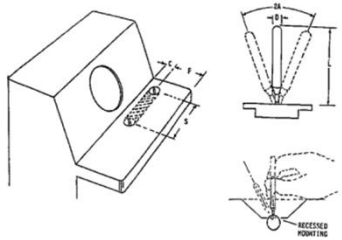
5.1.3.5.2.5 Dimensions and clearance. Joystick dimensions and clearance shall be in accordance with the criteria on figure 5.



	Dimensions		Resistance	A, Displacement	Clearance		
	D, Diameter	L, Length			S, Display CL to Stick CL	C, Around Stick	F, Stick CL to Shelf Front
Minimum	6.5 mm (0.25 in)	75 mm (3 in)	3.3 N (12 oz)		0	1/	120 mm (4.75 in)
Maximum	16 mm (0.625 in)	150 mm (6 in)	8.9 N (32 oz)	45° (785 mrad)	400 mm (15.75 in)		250 mm (10 in)

NOTE:
1/ Maximum stick excursion plus 100 mm (4 in).

FIGURE 5. Displacement joysticks.

5.1.3.5.2.5.1 Hand grip length. The hand grip length shall be 110 to 180 mm (4.3 to 7.1 in).

5.1.3.5.2.5.2 Hand grip diameter. The hand grip diameter shall be not more than 50 mm (2 in).

5.1.3.5.2.5.3 Clearance. Clearances of 100 mm (4 in) to the side and 50 mm (2 in) to the rear shall be provided to allow for hand movement.

5.1.3.5.2.5.4 Full range of motion. Clearance shall be provided to allow for the full range of joystick motion.

5.1.3.5.2.5.5 Actuation of buttons. Clearance shall be provided to enable actuation of joystick buttons while in any position within the joystick range of motion.

5.1.3.5.2.5.6 Mounting. Joysticks shall be mounted to provide forearm or wrist support.

5.1.3.5.2.5.7 Modular devices. Modular devices shall be mounted to allow actuation of the joystick without slippage, movement, or tilting of the mounting base.

5.1.3.5.3 Finger-operated displacement joysticks.

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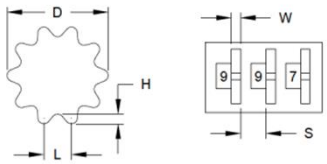

	D, Diameter	L, Trough Distance	W, Width	H, Depth	S, Separation	Resistance
Minimum	29 mm (1.125 in)	11 mm (0.43 in)	3 mm (0.125 in)	3 mm (0.125 in)	10 mm (0.4 in)	1.7 N (6 oz)
Maximum	75 mm (3 in)	19 mm (0.75 in)	--	6 mm (0.25 in)	--	5.6 N (20 oz)

FIGURE 8. Discrete thumbwheel control.

5.1.4.1.1.3.11 Resistance. Detents shall be provided for discrete position thumbwheels.

5.1.4.1.1.3.11.1 Elastic. Resistance shall be elastic, build up, and then decrease as each detent is approached so that the control snaps into position without stopping between adjacent detents.

5.1.4.1.1.3.11.2 Separation. The separation between adjacent edges of thumbwheel controls shall conform to the criteria on figure 8. Separation shall be sufficient to preclude accidental actuation of adjacent controls during normal control operation.

5.1.4.1.2 Continuous adjustment rotary controls.

5.1.4.1.2.1 Knobs. Knob design shall meet the criteria in 5.1.4.1.2.1.1 through 5.1.4.1.2.1.7.3.

5.1.4.1.2.1.1 Use. Knobs should be used when low forces or precise adjustments of a continuous variable are required. For most tasks, a moving knob with fixed scale should be used instead of a moving scale with fixed index. Special knob designs consisting of a bar-type knob and open-window skirt may be used for applications in which visual confusion may be reduced by exposing only one number of a scale at a time. Handle-like rotary controls may be used for special applications (i.e., a motorcycle accelerator control).

5.1.4.1.2.1.2 Single revolution controls. If positions of single revolution controls must be distinguished, a pointer or marker shall be available on the knob.

5.1.4.1.2.1.3 Shapes. Knobs that perform the same function shall have the same shape.

5.1.4.1.2.1.3.1 Knob function. Knob shape shall be determined by the knob's function and use.

5.1.4.1.2.1.3.2 Representative knob shapes. Representative knob shapes are shown on figure 9.

Applying Standards in Practice

Systematic process for selecting, interpreting, and verifying ergonomic standards and checklists throughout development

1 **Select Standards** >

Identification
Identify applicable ISO, ANSI, or industry-specific standards based on product type, user population, and context of use.

2 **Interpret Requirements** >

Translation
Translate technical standard clauses (e.g., "force < 25N") into specific, actionable design requirements for the project.

3 **Implement Design** >

Application
Apply requirements during concept and detailed design phases to ensure compliance is built-in from the start.

4 **Verify Compliance** >

Evaluation
Test prototypes against standard metrics to validate that the design meets the specified ergonomic criteria.